Chemistry 30 — Unit II Lesson 1
Thermochemical Changes: Foundations of Thermochemistry

What you will be able to do

By the end of this lesson, you will understand how chemical reactions involve energy changes, how energy is transferred between a system and its surroundings, and how these ideas apply to real processes such as photosynthesis, cellular respiration, and combustion.

Big idea

Thermochemistry examines how energy is absorbed or released during chemical reactions. Understanding energy flow allows chemists to explain why reactions occur and how they impact the environment.

1. Foundations of Thermochemistry

Every chemical reaction involves energy. Even when no obvious temperature change is felt, energy is still being transferred at the particle level. Thermochemistry provides the language and tools needed to describe these energy changes clearly and quantitatively.

Students often think of reactions only in terms of reactants and products, but this is incomplete. A reaction is better understood as an interaction between a chemical system and its surroundings. Recognizing this distinction is essential for understanding energy flow.

Key distinction

The system is the part of the universe being studied (usually the reacting chemicals). The surroundings include everything else, such as the container, air, and environment.

Energy can move between the system and the surroundings. When energy leaves the system, the surroundings gain energy and often feel warmer. When energy enters the system, the surroundings lose energy and may cool.

Important

A common misconception is that energy is used up during a reaction. Energy is never destroyed; it is transferred. Thermochemistry focuses on tracking this transfer accurately.

Worked example

Consider a combustion reaction occurring in an open container. As the fuel burns, heat is released. The reacting chemicals are the system, while the surrounding air and container are the surroundings. The warmth you feel indicates energy transfer from the system to the surroundings.

This reasoning is foundational for later lessons, where you will calculate exact energy values associated with reactions using enthalpy data.

Chemistry 30 — Unit II Lesson 1
Thermochemical Changes: Part B — Endothermic vs Exothermic Reactions

What you will be able to do

By the end of this section, you will be able to classify chemical reactions as endothermic or exothermic, explain the direction of energy flow during a reaction, and connect these ideas to real-world processes such as photosynthesis, cellular respiration, and hydrocarbon combustion.

Big idea

Chemical reactions are classified as endothermic or exothermic based on the direction of energy transfer between the system and its surroundings.

1B. Endothermic and exothermic reactions

Every chemical reaction involves a rearrangement of particles, and that rearrangement always involves energy. Some reactions release energy into their surroundings, while others require energy input to proceed. Thermochemistry provides a way to describe and classify these differences clearly.

Students often attempt to identify reactions as endothermic or exothermic by memorizing examples. A more reliable approach is to focus on the direction of energy flow. Understanding where the energy goes eliminates confusion when unfamiliar reactions are encountered.

Exothermic reactions

An exothermic reaction is one in which energy is released from the system to the surroundings. As a result, the surroundings experience an increase in temperature. This is why exothermic reactions are often described as reactions that feel warm or hot.

Hydrocarbon combustion is a common example of an exothermic process. When fuels such as gasoline or natural gas burn, chemical potential energy stored in the bonds is released as thermal energy, warming the surrounding environment.

Endothermic reactions

An endothermic reaction is one in which energy is absorbed by the system from the surroundings. Because energy is drawn in, the surroundings may cool as the reaction proceeds.

Photosynthesis is a classic example of an endothermic process. Plants absorb energy from sunlight to convert carbon dioxide and water into glucose. Without a continuous energy input, this reaction would not occur.

Key distinction

Exothermic reactions transfer energy to the surroundings, while endothermic reactions absorb energy from the surroundings.

Important

A common misconception is that endothermic reactions are cold and exothermic reactions are hot. In reality, these terms describe energy flow, not temperature alone. The temperature change observed depends on the surroundings and the conditions of the reaction.

Worked example

Cellular respiration is an exothermic process. As glucose reacts with oxygen, energy is released. This energy helps maintain body temperature in living organisms, demonstrating how chemical energy changes directly support biological systems.

Connection to Chemistry 30 thermochemistry

Classifying reactions as endothermic or exothermic is essential for later work with enthalpy changes (ΔH). In upcoming sections, you will learn how to calculate the magnitude of these energy changes using experimental data and thermochemical equations.

Chemistry 30 — Unit II Lesson 1

Part C: Energy Diagrams & Energy Profiles

What you will be able to do

In this section, you will learn how energy changes during a reaction can be visualized using energy diagrams. You will interpret energy profiles to distinguish between endothermic and exothermic reactions, identify activation energy, and explain why some reactions require an initial energy input even when they ultimately release energy.

Big Idea

Energy diagrams provide a visual model that connects particle collisions, bond energy changes, and observed temperature effects. They help explain not just whether energy is absorbed or released, but why reactions occur at different rates.

1. What an Energy Diagram Represents

An energy diagram is a graph that shows how the potential energy of a system changes as a chemical reaction proceeds. The vertical axis represents potential energy, while the horizontal axis represents the progress of the reaction. This horizontal axis is not time; instead, it tracks how bonds are breaking and forming.

Students often assume that reactions happen smoothly from reactants to products. In reality, bonds must first be broken, and breaking bonds always requires energy. This initial energy requirement explains why even exothermic reactions do not begin spontaneously without a trigger.

Activation Energy (E_a)

Activation energy is the minimum amount of energy that reacting particles must have in order to form an activated complex. The activated complex is a temporary, unstable arrangement of atoms that exists at the peak of the energy diagram.

A common misconception is that activation energy is 'used up' during the reaction. In fact, this energy is temporarily stored as bonds stretch and rearrange, then released again as new bonds form.

2. Energy Diagrams for Exothermic Reactions

In an exothermic reaction, the products have lower potential energy than the reactants. This difference in energy corresponds to a negative ΔH value. The excess energy is released to the surroundings, often as heat.

3. Energy Diagrams for Endothermic Reactions

In an endothermic reaction, the products have higher potential energy than the reactants. The reaction absorbs energy from the surroundings, resulting in a positive ΔH value. Energy diagrams make this difference immediately visible.

Worked Example

Consider two reactions with identical activation energies, but different overall energy changes. Reaction A releases 120 kJ of energy, while Reaction B absorbs 85 kJ. On an energy diagram, Reaction A would show products far below reactants, while Reaction B would show products above reactants.

This visual difference reinforces why ΔH depends on the relative energies of reactants and products, not on the height of the activation energy barrier.

Connection to Thermochemistry Calculations

Later in this unit, you will calculate ΔH values numerically. Energy diagrams help you predict the sign of ΔH before performing calculations. They also explain why catalysts lower activation energy without changing ΔH, a concept that becomes essential when comparing reaction pathways.

Chemistry 30 — Unit II Thermochemical Changes
Lesson 1 — Part D: Endothermic vs Exothermic Systems

What you will be able to do

By the end of this section, you will be able to clearly distinguish between endothermic and exothermic systems, interpret energy changes using ΔH notation, and explain these processes using particle-level reasoning.

Big Idea

Chemical reactions do not create or destroy energy. Instead, energy is transferred between the reacting system and its surroundings. Whether a reaction feels hot or cold depends on the *direction* of that energy transfer.

Understanding Energy Transfer

In thermochemistry, the system refers to the chemicals involved in the reaction, while the surroundings include everything else—containers, air, and nearby materials. Energy can move between these two regions, but it is never lost.

Exothermic Systems

An exothermic reaction releases energy from the system into the surroundings. As chemical bonds rearrange, the products end up at a lower potential energy than the reactants. The excess energy appears as heat, light, or sound.

Endothermic Systems

An endothermic reaction absorbs energy from the surroundings into the system. In this case, more energy is required to break bonds than is released when new bonds form, leaving the products at a higher potential energy level.

Common Student Confusion

A reaction that feels cold is not 'lacking energy.' Instead, it is actively pulling energy from the surroundings. This often surprises students because the temperature change is observed outside the system, not within it.

Connection to Thermochemistry Calculations

In Chemistry 30, energy changes are quantified using enthalpy (ΔH). A negative ΔH value indicates an exothermic process, while a positive ΔH value indicates an endothermic process. These signs become critically important in later calculations.

Chemistry 30 — Unit II Lesson 1
Part E: Energy Diagrams & Potential Energy

What you will be able to do

By the end of this section, you will be able to interpret energy diagrams, explain how potential energy changes during a reaction, identify activation energy and enthalpy change (ΔH), and connect these graphical models to real chemical systems.

Big idea

Energy diagrams visually represent how potential energy changes as reactants are converted into products. They help explain why reactions require an initial input of energy and why some reactions release more energy than others.

1E. Why energy diagrams are used

Energy diagrams are models that help chemists visualize changes in potential energy during a chemical reaction. Although energy itself cannot be seen directly, its effects can be represented graphically to improve understanding.

Students often expect reactions to proceed smoothly from reactants to products. In reality, bonds must be broken before new ones can form, and this process always requires energy. Energy diagrams show this requirement clearly.

Potential energy in chemical systems

Potential energy in chemistry refers to the energy stored in chemical bonds. Stronger bonds generally store less potential energy than weaker bonds. During a reaction, energy is absorbed to break bonds and released when new bonds form.

The relative heights of reactants and products on an energy diagram reflect their potential energies. The difference between these levels corresponds to the enthalpy change (ΔH) of the reaction.

Activation energy

Activation energy is the minimum energy required for a reaction to occur. It represents the energy needed to reach the activated complex, a temporary and unstable arrangement of atoms.

Activation energy explains why many reactions do not occur spontaneously at room temperature.

Even highly exothermic reactions require activation energy. This is why fuels such as gasoline need a spark to ignite.

Comparing exothermic and endothermic diagrams

In an exothermic reaction, products are at a lower potential energy than reactants, resulting in a negative ΔH value. In an endothermic reaction, products are at a higher potential energy, resulting in a positive ΔH value.

Energy diagrams make this difference immediately visible, allowing chemists to predict energy changes before performing calculations.

Worked example

Consider two reactions with identical activation energies. Reaction A has products far below the reactants, while Reaction B has products above the reactants. Reaction A is exothermic, while Reaction B is endothermic.

Connection to Chemistry 30 thermochemistry

Energy diagrams will be used repeatedly in Unit II to interpret thermochemical equations, compare reaction pathways, and understand how catalysts affect reactions without changing ΔH.

Chemistry 30 — Unit II Lesson 1
Part F: Catalysts, Reaction Pathways & Energy Efficiency

What you will be able to do

By the end of this section, you will be able to explain how catalysts affect reaction pathways, interpret how activation energy can be lowered without changing enthalpy (ΔH), and connect these ideas to real-world applications such as industrial efficiency, environmental impact, and biological systems.

Big idea

Catalysts increase reaction rates by providing an alternative reaction pathway with a lower activation energy, without being consumed and without changing the overall energy change of the reaction.

1F. Why catalysts matter

Many chemical reactions are thermodynamically favourable but proceed too slowly to be useful. Catalysts allow these reactions to occur at practical rates by lowering the energy barrier that must be overcome.

Students often assume that catalysts add energy to a reaction. In reality, catalysts change the pathway by which reactants are converted to products, reducing the activation energy required.

Reaction pathways and activation energy

An uncatalyzed reaction follows a pathway with a relatively high activation energy. When a catalyst is present, the reaction follows a different pathway with a lower activation energy.

Catalysts do not change the potential energies of reactants or products, and therefore do not affect ΔH.

Real-world example: industrial catalysis

The Haber process, used to synthesize ammonia for fertilizers, relies on iron-based catalysts. Without a catalyst, the reaction would proceed far too slowly to support global food production.

Catalysts in biological systems

Enzymes are biological catalysts that allow essential reactions to occur rapidly at body temperature. Without enzymes, many life-sustaining reactions would not proceed fast enough to support life.

Important

A common misconception is that catalysts make reactions release more energy. Catalysts only affect reaction rate, not the total energy change.

Connection to Chemistry 30 thermochemistry

Understanding catalysts prepares you to evaluate energy efficiency in industrial processes. Later in Unit II, you will analyze energy changes in systems and consider how catalysts help reduce energy consumption and environmental impact.

Chemistry 30 – Thermochemical Changes

Lesson 1: Practice & Check-for-Understanding

This practice set is designed to help you check your understanding of the core ideas from Lesson 1 before moving on to calculations. If you can explain your reasoning clearly for these questions, you are well prepared for quantitative thermochemistry.

Section A — Core Concepts (Explain in Words)

1. Explain, in your own words, what is meant by the term system in thermochemistry. How does it differ from surroundings?

2. Why is energy considered to be transferred rather than created or destroyed during a chemical reaction?

3. Describe how temperature is related to particle motion using the kinetic molecular theory.

4. Explain why a reaction can release energy without becoming hotter.

5. Describe the difference between potential energy and kinetic energy using a chemical example.

Section B — Energy Diagrams & Reaction Pathways

6. Sketch and label a potential energy diagram for an exothermic reaction. Identify reactants, products, activation energy, and ΔH.

7. How does the height of the activation energy barrier affect reaction rate?

8. Explain why catalysts lower activation energy but do not change ΔH.

9. Describe how energy diagrams help chemists compare different reaction pathways.

Section C — Real-World & STS Connections

10. Explain why catalysts are critical in industrial processes such as fertilizer production.

11. Describe one way thermochemistry influences energy efficiency or sustainability.

12. Explain why understanding energy changes is important when evaluating alternative fuels.

Section D — Self-Check (Ready to Move On?)

You should be able to confidently answer YES to all of the following:

- I can explain energy transfer without using formulas.
- I can interpret and explain energy diagrams.
- I understand the role of activation energy and catalysts.
- I can connect thermochemistry to real-world energy systems.

Chemistry 30 – Thermochemistry Unit
Lesson 1 Practice: Teacher Answer Key

Part A – Energy and Systems

Energy is defined as the capacity to do work or produce change. In thermochemistry, we focus on energy transfers between a system and its surroundings. Students should emphasize that energy is conserved and transferred, not created or destroyed. Common misconceptions include thinking that energy is 'used up' rather than redistributed.

Part B – Temperature vs Energy

Temperature measures the average kinetic energy of particles, not total energy. A small object at a high temperature can contain less total energy than a large object at a lower temperature. Students often incorrectly assume temperature and energy are interchangeable; reinforce the distinction.

Part C – Heat Transfer

Heat is energy transferred due to a temperature difference. Students should clearly identify direction: heat flows from warmer to cooler objects. Language such as 'cold flows' should be corrected early.

Part D – Endothermic vs Exothermic

Endothermic processes absorb energy from the surroundings, causing the surroundings to cool. Exothermic processes release energy, warming the surroundings. Students should link this to observable temperature changes rather than memorized definitions.

Part E – Energy Diagrams

Energy diagrams visually represent potential energy changes. For exothermic reactions, products lie lower than reactants; for endothermic reactions, products lie higher. Activation energy represents the minimum energy required for reaction to occur.

Part F – Big Picture Check

Students should articulate that thermochemistry explains why reactions feel hot or cold and how energy relates to chemical change. Connections to real-world contexts such as heating packs, combustion, and metabolic energy should be encouraged.